llmXive Automated Review of Kairos: A Native World Model Stack for Physical AI

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong factual claims regarding benchmark performance and theoretical guarantees that require verification against the cited sources and internal consistency.

First, the claim in Section 4.1 (e002) and Table 1 that Kairos achieves "perfect scores" (1.00) in Newtonian mechanics, fluid dynamics, and gravity on WorldModelBench is highly suspect. In empirical benchmarks, especially those involving physical commonsense, achieving a mathematically perfect 1.00 across multiple distinct sub-domains is statistically rare and suggests either a rounding artifact or a potential overstatement of results. If these values are rounded from 0.995+, the text should reflect this (e.g., "near-perfect" or "0.99+") to maintain scientific rigor.

Second, the citation of "Qwen3.5" and "Qwen3" as established models used for the VLM backbone (e.g., e000, e002) and captioning (e001) relies on future-dated or non-public preprints (BibTeX lists Qwen3.5 as Feb 2026). While the paper is a preprint, the specific benchmark scores attributed to these models in Table 3 (e.g., MMMU 64.2, MathVista 76.7) must be verifiable. If these models are not yet public, the authors must clarify if these are internal evaluations or if the citations are placeholders for future work, as using unreleased models as baselines for comparison can be misleading.

Third, the claim of "linear scaling in inference time" (Abstract, e000) is supported by the latency table (e001) and Figure 1c, but the theoretical section (Theorem 2, e002) provides a bound on *excess risk* ($\mathcal{R}_t - \mathcal{R}_t^\star$) based on memory contraction, not computational complexity ($O(N)$). While the architecture (GLA) is designed for linear complexity, the text should carefully distinguish between the *statistical* guarantee of long-horizon state maintenance (proven in the theorems) and the *algorithmic* efficiency (empirically shown). Conflating "sufficiency of memory" with "linear time complexity" in the narrative could confuse readers about what the theorems actually prove.

Finally, the speedup claims in Table 2 (e001) range from 28x to 85x. The calculation for 1GPU (2526/43   58x) and 4GPU (687/9   76x) falls within this range, but the lower bound of 28x is not immediately obvious from the provided table rows. The authors should specify which baseline configuration or metric yields the 28x figure to ensure the claim is fully supported by the presented data.

Action items.

- [**writing**] The claim of 'perfect scores' (1.00) in Newtonian, fluid, and gravity on WorldModelBench (Table 1, e002) is statistically improbable. Verify if these are rounded values; if so, rephrase to 'near-perfect' or provide exact decimals to avoid misleading readers.
- [**writing**] The latency speedup range of 28x–85x (Table 2, e001) lacks a clear derivation for the 28x lower bound from the provided 1GPU/4GPU data. Clarify which specific configuration yields the 28x figure or adjust the range to match the calculated ratios (~58x–76x).
- [**science**] The paper cites 'Qwen3.5' (2026 preprint) and attributes specific benchmark scores (e.g., MMMU 64.2) to it (Table 3, e002). Verify these models are publicly available and scores are accurate, as using unreleased models as baselines can be misleading.

- **[writing]** The text conflates the theoretical guarantee of 'state propagation' (risk bounds in Theorem 2) with 'linear time complexity' (algorithmic efficiency). Explicitly distinguish between the statistical sufficiency of the memory and the computational complexity claims.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 presents a Sankey diagram of the data filtering pipeline but fails to include a legend for color coding or any quantitative values on the flows, making the visualization ambiguous and lacking the necessary data to support pipeline claims.

Figure 2

Figure 2 effectively displays representative examples from the dataset across four distinct categories (General, Human, Physics, Robot). The column headers clearly identify the domains, and the visual content is high-quality and legible, matching the caption's description.

Figure 3

Figure 3 effectively illustrates the 'Divide-and-Conquer' captioning strategy by visually contrasting a basic VLM-generated caption (green box) with a more detailed, tag-enhanced version (blue box). The layout is clear, the text is legible, and the figure aligns perfectly with its caption's description of the comparison.

Figure 4

Figure 4 effectively illustrates the paper's 'Enhanced Text with Chain of Thought' concept by visually contrasting physics-centric captions with long-term task captions. The layout is clear, the text is legible, and the content aligns perfectly with the provided caption description.

Figure 5

The figure displays a grid of robot simulation frames consistent with the dataset name, but the caption contains a spelling error ('WordelModelBench').

Figure 6

Figure 6 presents a grid of representative video frames from the DreamGen dataset, illustrating diverse robotic manipulation tasks. The visual content is clear, and the caption accurately describes the figure as samples from the specified dataset.

Figure 7

Figure 7 presents a grid of video frames showing robot manipulation tasks, consistent with the caption's description of 'Kairos samples on the Paibench robot dataset.' The figure is clear, legible, and effectively communicates the visual content of the dataset without requiring additional legends or axes.

Figure 8

The figure effectively communicates human evaluation results with clear data labels and a legend, but the caption is overly brief and the repetitive y-axis labels across subplots create unnecessary visual redundancy.

Figure 9

Figure 9 displays input and predicted frames from the PAI-Bench dataset but lacks explicit description in the caption of the temporal structure (number of predicted frames) and the meaning of the column headers.

Figure 10

The figure effectively displays video generation results with clear internal labels, but the caption fails to explicitly define the layout (input vs. predicted frames) or the meaning of the 'TI2V' acronym, relying on visual inference.

Figure 11

Figure 11 effectively displays text-to-video generation samples from the WorldModelBench dataset. The layout clearly pairs descriptive prompts on the left with the corresponding generated video frames on the right, demonstrating the model's ability to render complex physical scenes and actions.

Figure 12

The figure displays generated video frames but suffers from visible temporal artifacts and lacks a descriptive caption to contextualize the visual content or the dataset.

Action items.

- [**science**] Figure 1: The Sankey diagram lacks a legend defining the color coding (e.g., purple, beige, yellow, green, grey) for the nodes and flows, making it impossible to distinguish between data sources, intermediate stages, and specific filter types.

- [**science**] Figure 1: The diagram is missing quantitative units or values (e.g., number of samples, percentages) on the flows or nodes, rendering the 'pipeline' visualization qualitative and unable to support claims about data volume reduction or filtering efficiency.
- [**writing**] Figure 5: The caption contains a typo, spelling the dataset as 'WordelModelBench' instead of 'WorldModelBench'.
- [**writing**] Figure 8: The caption 'Human evaluation results' is too generic for a complex multi-panel figure; it should specify that the figure compares Kairos against four specific baselines (cosmos-predict2.5, lingbot-14B, wan2.2) across three datasets.
- [**writing**] Figure 8: The y-axis labels 'Paibench robot-subset', 'WorldModelBench robot-subset', and 'Dreamgen' are repeated in every subplot; consolidating these into a single shared axis or using a grid layout would reduce visual clutter.
- [**writing**] Figure 9: The caption states 'Kairos samples on the PAI-Bench dataset' but does not specify that the images show 'Input frame' followed by 'Predicted frames', making the temporal nature of the samples unclear.
- [**writing**] Figure 9: The column headers 'Input frame' and 'Predicted frames' are present but the number of predicted frames per input is not specified in the caption, leaving the reader to infer the temporal span.
- [**writing**] Figure 10: The caption 'Kairos samples(TI2V)' is ambiguous; it does not explicitly state that the left column represents the input frame and the subsequent columns represent the predicted frames, forcing the reader to infer the temporal progression.
- [**writing**] Figure 10: The image contains internal labels ('input frame', 'Predicted frames') that are not referenced or defined in the caption, creating a disconnect between the visual annotation and the textual description.
- [**science**] Figure 12: The 'Generated Frames' columns show significant visual artifacts and temporal inconsistencies (e.g., flickering mixer, morphing coffee stream, unstable waterfall) compared to the 'Prompt' column, which undermines the claim of generating high-quality physical AI samples.
- [**writing**] Figure 12: The caption is overly brief and does not explain the layout (e.g., that the first column is the prompt and subsequent columns are generated frames) or the nature of the 'VideoPhy dataset' samples shown.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript suffers from significant jargon overuse, relying heavily on unexplained acronyms and marketing-style terminology that obscures meaning for non-specialist readers. The term "Physical AI" is used repeatedly as a proper noun without definition, creating an immediate barrier to entry. Similarly, "Native" is used as a modifier for "Pre-training Paradigm" and "Unified Architecture" without technical justification, functioning more as a buzzword than a descriptor.

Acronyms are frequently introduced without definition. "CEDC," "SOTA," "WAM," "DiT," "GLA," "SWA," "DSWA," "MoT," "DPO," "SFT," "VAE," "CoT," "DMD," and "CM" all appear in the text or tables before being spelled out. For instance, in Section 2.1, the text lists "SWA, DSWA, GLA" without defining them, and later in Section 5.1, "DMD" and "CM" are used without prior expansion. Table 1 relies entirely on acronyms like "T2I" and "N2V" which are only defined in footnotes, forcing the reader to cross-reference constantly.

Furthermore, the paper uses dense technical shorthand like "FP8," "INT8," and "INT4" in Section 5.2 without clarifying the bit-depth or format for a general audience. The consistent failure to define these terms at their first occurrence violates standard academic writing practices and unnecessarily excludes readers who are not deeply embedded in the specific sub-fields of diffusion models and robotics. The text should be revised to spell out all acronyms upon first use and replace vague marketing terms like "Native" with precise technical descriptions.

Action items.

- [**writing**] The manuscript suffers from significant jargon overuse, relying heavily on unexplained acronyms and marketing-style terminology that obscures meaning for non-specialist readers. The term "Physical AI" is used repeatedly as a proper noun without definition, creating an immediate barrier to entry. Similarly, "Native" is used as a modifier for "Pre-training Paradigm" and "Unified Architecture" without technical justification, functioning more as a buzzword than a descriptor. Acronyms are frequently

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The logical consistency of the paper is generally strong in its theoretical framing but contains gaps in the causal linkage between the proposed mechanisms and the empirical claims.

First, the theoretical argument for the necessity of persistent states (Theorem 1, Section 3.3 and Section 6.1) is logically sound: it correctly identifies that if a target depends on history outside a window, a window-restricted predictor incurs irreducible excess risk. However, the paper asserts that baseline models (Cosmos, Wan) fail on long-horizon tasks *because* of this theoretical limitation. This causal claim is not fully supported. The paper does not present an ablation study where the window size of a baseline is varied to demonstrate the predicted excess risk, nor does it show that the specific failure modes (e.g., object permanence loss) correlate with the theoretical conditions derived. The connection between the abstract theorem and the specific empirical failures of competitors is asserted rather than demonstrated.

Second, the claim of “mathematically guaranteeing state propagation” (Abstract) rests on the contractive property of the Gated Linear Attention (GLA) module (Theorem 2). The theorem states that if the global memory update is contractive ($\rho < 1$), the error is bounded. While the architecture is designed to be contractive, the paper does not provide empirical verification that the trained model actually satisfies $\rho < 1$ across the training trajectory. Without this verification, the “guarantee” is conditional on an unverified parameter state, weakening the logical force of the claim.

Third, the claim of “linear scaling” in inference time (Abstract, Figure 1c) is supported by latency tables showing linear growth with resolution/frames. However, the paper attributes this primarily to the Hybrid Linear Attention mechanism. This causal attribution is confounded by other efficiency techniques mentioned, such as 4-step distillation, FP8 quantization, and operator fusion. The paper does not isolate the contribution of the attention mechanism to the scaling behavior. It is logically possible that the scaling is linear due to the distillation (reducing steps) or hardware optimizations, while the attention mechanism’s complexity reduction is secondary. The current presentation implies the attention mechanism is the primary driver without sufficient evidence to rule out the other factors.

Finally, the definition of “Native” in the title and abstract implies a unified, end-to-end approach. The logical consistency of this claim is slightly undermined by the modular description of the training pipeline (Phase I, II, III) which appears to be a sequential curriculum rather than a truly simultaneous, native joint optimization of all components from the start. While the “Joint World-Action Training” (Stage III) addresses this, the distinction between “curriculum” and “native” is not rigorously defined, leading to a slight ambiguity in the core contribution’s definition.

Action items.

- [**writing**] The logical consistency of the paper is generally strong in its theoretical framing but contains gaps in the causal linkage between the proposed mechanisms and the empirical claims. First, the theoretical argument for the necessity of persistent states (Theorem 1, Section 3.3 and Section 6.1) is logically sound: it correctly identifies that if a target depends on history outside a window, a window-restricted predictor incurs irreducible excess risk. However, the paper asserts that baseline model

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper exhibits significant overreach in its framing of theoretical guarantees and performance claims.

First, the Abstract and Conclusion repeatedly assert that the architecture "mathematically guarantee[s] state propagation" and "guarantee[s] long-horizon state maintenance." This is a strong over-claim. The theoretical section (Section 6) provides theorems establishing the *necessity* of persistent states for supra-window dependence and the *approximate sufficiency* of a hybrid memory under specific conditions (e.g., contractive global memory, Lipschitz continuity, factorized predictors). These theorems prove that *if* the model satisfies these conditions, the risk is bounded. They do not prove that the specific Kairos implementation *guarantees* these conditions hold in practice, nor do they guarantee zero error or perfect state propagation. The language should be revised to reflect that the architecture is *motivated by* and *designed to satisfy* these theoretical bounds, rather than claiming a mathematical guarantee of the outcome.

Second, the claim of achieving "SOTA" (State-of-the-Art) across "diverse benchmarks" (Figure 1 caption, Abstract) is over-extended. The evaluation is limited to a specific set of benchmarks: WorldModelBench, DreamGen, PAI-Bench, LIBERO-Plus, and RoboTwin 2.0. While Kairos performs well on these, the paper does not provide evidence that it is SOTA on the broader landscape of world model or video generation benchmarks (e.g., VBench, UCF101, specific autonomous driving simulators). The claim should be qualified to "SOTA on the evaluated embodied and long-horizon benchmarks" to accurately reflect the scope of the evidence.

Third, the Conclusion claims Kairos sustains "true real-time, closed-loop inference on consumer-grade edge hardware." The data in Table 3 shows a latency of 5.7 seconds on an RTX 5090 for generating a 5-second video clip at 480p. This is not "real-time" in the context of a closed-loop control system, which typically requires inference latencies in the range of milliseconds (e.g., <33ms for 30fps). A 5.7-second latency for a 5-second horizon implies a non-causal or highly delayed loop, contradicting the "real-time" assertion. The term "real-time" is misleading here and overstates the deployment capabilities demonstrated by the reported latency figures.

Action items.

- **[writing]** The paper exhibits significant overreach in its framing of theoretical guarantees and performance claims. First, the Abstract and Conclusion repeatedly assert that the architecture "mathematically guarantee[s] state propagation" and "guarantee[s] long-horizon state maintenance." This is a strong over-claim. The theoretical section (Section 6) provides theorems establishing the *necessity* of persistent states for supra-window dependence and the *approximate sufficiency* of a hybrid memory under

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a technically ambitious framework for Physical AI but raises significant safety and ethical concerns that require clarification before the work can be considered for deployment or further publication.

Data Privacy and Consent (Section Data Collection): The authors state they employ a “hybrid strategy” involving “internet-crawled” data and “first-person human manipulation” videos, accumulating “hundreds of millions of standardized clips” (Section Data, Subsection Data Collection). While the paper mentions filtering for NSFW content, it fails to address the legal and ethical provenance of this data. Specifically, there is no mention of:

1. **Consent:** How was consent obtained from individuals appearing in the “human-centric” and “first-person” videos?
2. **Copyright:** What is the licensing status of the “internet-crawled” material?
3. **Privacy:** How are personally identifiable information (PII) and biometric data (faces, voices) handled in the training set?

Without a clear statement on compliance with GDPR, CCPA, or similar frameworks, the use of such a massive, uncurated dataset poses a severe privacy risk.

Safety of Generated Outputs (Section Evaluation Results): The paper claims state-of-the-art performance on physical benchmarks (WorldModelBench, PAI-Bench) but lacks a dedicated safety evaluation. For a model designed to control physical agents (“Physical AI”), the ability to generate physically plausible but *unsafe* actions (e.g., dropping heavy objects on humans, manipulating hazardous materials) is a critical failure mode. The current evaluation focuses on “Physics Adherence” and “Instruction Following” but does not explicitly test for safety constraints or adversarial robustness against harmful prompts. A “Safety Evaluation” subsection detailing red-teaming results is mandatory.

Autonomous Self-Evolution Risks (Section Inference): The “Self-evolution” and “Prompt Self-alignment” sections describe a closed-loop system where the model autonomously rewrites prompts and refines its own policies based on internal reward functions without human intervention. This introduces significant risks of:

1. **Reward Hacking:** The model might optimize for the internal reward function in ways that violate safety constraints.
2. **Drift:** Unsupervised self-improvement could lead to the emergence of unsafe behaviors over time.

The paper must discuss the safeguards, monitoring mechanisms, and human-in-the-loop requirements for these autonomous loops.

Dual-Use Concerns: The capability to simulate complex physical interactions and generate high-fidelity robotic control policies has inherent dual-use potential. The authors should include a brief discussion on the responsible release of the model and data, potentially considering access controls or usage agreements to prevent misuse in malicious physical applications.

Action items.

- **[writing]** Consent: How was consent obtained from individuals appearing in the “human-centric” and “first-person” videos?

- [**writing**] Copyright: What is the licensing status of the "internet-crawled" material?
- [**writing**] Privacy: How are personally identifiable information (PII) and biometric data (faces, voices) handled in the training set? Without a clear statement on compliance with GDPR, CCPA, or similar frameworks, the use of such a massive, uncurated dataset poses a severe privacy risk. Safety of Generated Outputs (Section Evaluation Results): The paper claims state-of-the-art performance on physical benchmarks (WorldModelBench, PAI-Bench) but lacks a dedicated safety evaluation. For a model designed to co
- [**writing**] Reward Hacking: The model might optimize for the internal reward function in ways that violate safety constraints.
- [**writing**] Drift: Unsupervised self-improvement could lead to the emergence of unsafe behaviors over time. The paper must discuss the safeguards, monitoring mechanisms, and human-in-the-loop requirements for these autonomous loops. Dual-Use Concerns: The capability to simulate complex physical interactions and generate high-fidelity robotic control policies has inherent dual-use potential. The authors should include a brief discussion on the responsible release of the model and data, potentially considerin

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence supporting the central claims of the Kairos framework is currently insufficient to justify the strong assertions made in the abstract and conclusion.

First, the theoretical guarantees presented in Section 2.3 and Section 5 are not empirically validated. The paper claims that the Hybrid Linear Temporal Attention mechanism "mathematically guarantees state propagation" and that the hybrid memory is "approximately sufficient" (Theorem 2). These theorems rely on the assumption that the global memory update is contractive (i.e., the spectral radius $\rho < 1$). While the authors define the Gated Linear Attention (GLA) mechanism, they provide no empirical evidence that the *learned* parameters actually satisfy this contraction condition during training. Without measuring the spectral radius of the learned state matrices or providing a proof that the training objective enforces this constraint, the claim of a "guarantee" is unsupported. The "exact sufficiency" corollary (Corollary 3) assumes zero approximation error, a condition that is never met in practice and is not addressed in the experimental analysis.

Second, the empirical evidence for "State-of-the-Art" (SOTA) performance is weak due to a lack of statistical rigor. In Table 1 (WorldModelBench), Kairos scores 9.30 compared to Cosmos3-Nano's 9.26. In Table 2 (DreamGen Bench), the differences are similarly marginal. The paper presents single-point estimates without reporting standard deviations, confidence intervals, or p-values derived from multiple random seeds. Given the small margins of victory, it is impossible to determine if these results are statistically significant or merely artifacts of random initialization or evaluation noise. The ablation studies (Table 3) suffer from the same issue; the reported gains from human-centric data scaling (9.08 to 9.25) are not accompanied by variance metrics, making it difficult to assess the reliability of the contribution.

Third, the efficiency comparisons in Table 4 are potentially misleading. The paper claims a 28x–85x speedup over baselines like Cosmos-Predict2.5-14B. However, this comparison conflates model size (4B vs 14B) with architectural efficiency. A fair scientific comparison of "efficiency" should normalize for the number of parameters or the total FLOPs required for inference. Comparing a 4B model directly to a 14B model without controlling for scale does not prove that the *architecture* is more efficient, only that a smaller model is faster. The paper needs to demonstrate that Kairos achieves comparable performance to a same-scale baseline with significantly lower latency, or that it achieves higher performance at the same compute budget.

Finally, the long-horizon generation results (Table 5) show Kairos maintaining a score of 79.9 while baselines drop to ~77.2. While this suggests better stability, the absolute scores remain relatively low (below 80), and the paper does not analyze the specific types of failures (e.g., object permanence vs. motion smoothness) to validate the claim that the hybrid attention specifically solves the long-horizon problem. The evidence suggests a trend but does not robustly isolate the causal mechanism of the proposed architecture.

Action items.

- [**science**] The claim of 'mathematically guaranteeing state propagation' (Abstract) and 'exact sufficiency' (Corollary 3) relies on unverified assumptions about the contractivity of the learned Gated Linear Attention (GLA) gates. The paper must provide empirical evidence (e.g., spectral radius analysis of learned gates) or a rigorous proof that the learned parameters satisfy the contraction condition (rho $<$ 1) required by Theorem 2, rather than assuming it holds by design.

- [**science**] Benchmark results (e.g., Table 1, Table 2) show marginal or statistically insignificant differences between Kairos and baselines (e.g., 9.30 vs 9.26 on WorldModelBench). The paper lacks statistical significance testing (p-values, confidence intervals, or standard deviations over multiple seeds) to support the claim of 'SOTA' performance. Without this, the observed gains could be due to random variance.

- [**science**] The ablation study in Table 3 (Human-Centric Data Scaling) reports a Total Score increase from 9.08 to 9.25. However, the paper does not specify the number of training runs, seeds, or variance associated with these scores. Given the small absolute gain (~1.8%), the robustness of this claim is questionable without statistical validation.

- [**science**] The efficiency claims (Table 4) compare Kairos-4B against baselines with different parameter counts and architectures. The paper fails to normalize for compute budget (FLOPs) or training data volume. The '28x-85x speedup' is a raw latency comparison that does not account for the fact that baselines may be larger models; a fair comparison requires controlling for model scale or training compute.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical rigor of the experimental evaluation is insufficient to support the paper's central claims of state-of-the-art performance and theoretical guarantees. While the architecture and training pipeline are described in detail, the analysis of the results lacks fundamental statistical components required for a scientific publication in this domain.

First, the **Human Evaluation** section (Section 5.1.1) reports win rates (e.g., "60.2% vs Cosmos-Predict2.5") based on "10 volunteers." The manuscript fails to specify the statistical methodology used to derive these percentages. It is unclear if these are raw counts, if multiple trials were conducted per prompt, or what the confidence intervals are. A binomial test or a confidence interval (e.g., Wilson score interval) is necessary to determine if a 60.2% win rate is statistically distinguishable from a random 50% baseline, especially with a small sample size of evaluators. Without p-values or error margins, these claims are anecdotal.

Second, the **Ablation Studies** (Section 5.1.1 and 5.2.1) report performance improvements (e.g., "Instruction Following from 2.10 to 2.33") without providing the number of evaluation episodes (N) or the number of random seeds used for training. In stochastic environments like robotics (LIBERO-Plus, RoboTwin) and generative modeling, performance variance is high. Reporting a single point estimate without standard deviation or a statistical test (e.g., paired t-test) makes it impossible to assess the significance of the reported gains. A +0.23 improvement could easily be within the noise of the evaluation metric.

Third, the **Benchmark Results** tables (Tables 1, 2, 3, 6) present point estimates for complex metrics (e.g., "Total Score 9.30", "AVG_PA 0.538") without any measure of uncertainty. For diffusion models, generation quality varies significantly between runs. The absence of standard deviations or confidence intervals prevents a fair comparison with baselines that may have similar mean scores but different variances. The claim of "SOTA" is statistically weak without demonstrating that the observed differences are significant.

Finally, the **Theoretical Analysis** (Section 6) derives bounds on excess risk involving constants like Lipschitz constants (L) and contraction rates (ρ). However, the paper does not attempt to estimate these constants empirically or validate the bounds against the actual observed errors on the test set. The theoretical contribution remains abstract and unconnected to the empirical evidence, failing to demonstrate that the proposed architecture actually operates within the predicted theoretical limits.

To proceed, the authors must re-run evaluations with sufficient seeds to report mean $\pm$ standard deviation, perform statistical significance testing on human evaluations and ablation studies, and either validate the theoretical bounds empirically or clearly frame them as purely theoretical upper bounds without empirical verification.

Action items.

- **[science]** Human evaluation methodology is statistically under-specified. Section 'Human Evaluation' cites '10 volunteers' but omits the statistical test used (e.g., binomial test, t-test), confidence intervals, or p-values for the reported win rates (e.g., 60.2% vs Cosmos). Without variance estimates or significance testing, these claims are anecdotal rather than statistical evidence.
- **[science]** Ablation study sample sizes are missing. Tables 4 and 5 report performance deltas (e.g., +6.0 on LIBERO-Plus) but do not state the number of evaluation episodes or seeds used. Without N, it is impossible to determine if these gains are statistically significant or within the noise floor of the benchmark.
- **[science]** Benchmark metrics lack uncertainty quantification. Tables 1, 2, 3, and 6 present point estimates for scores (e.g., 9.30, 0.538) without standard deviations, confidence intervals, or error bars. Given the stochastic nature of diffusion models and RL, single-point reporting is insufficient to claim 'SOTA' status over baselines with similar scores.
- **[science]** The theoretical 'excess risk' bounds in Section 6 are not empirically validated. The paper derives bounds involving constants (L, rho, epsilon) but provides no empirical estimation of these parameters or a comparison between the theoretical bound and the observed error on the test set, rendering the theoretical contribution disconnected from the experimental results.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a comprehensive technical contribution, but the writing quality is currently compromised by significant structural redundancy and inconsistent formatting that impedes readability. The most critical issue is the duplication of major sections. Specifically, the "Conclusion and Future Works" and "Theoretical Analysis" sections appear in multiple chunks (e.g., e002 and e003) with slight variations in phrasing. This suggests a compilation error or a failure to merge draft versions, which confuses the narrative flow and forces the reader to reconcile conflicting statements. These sections must be consolidated into single, authoritative versions.

Additionally, the formatting of subsection headers in the pretraining paradigm (Section 3.2) relies on fragile LaTeX macros (`\uppercase\expandafter{\romannumeral ...}`) which may render poorly or inconsistently across different viewers. It is recommended to use standard text (e.g., "Stage II") for better clarity and robustness.

There is also unnecessary repetition in the "Data Engineering Infrastructure" section, where Table 2 and its accompanying text are duplicated from an earlier section. This redundancy disrupts the logical progression of the paper. Finally, while the figure captions generally describe the content, some (like Figure 3 in the Introduction) are split across subfigures with a summary that lacks specific detail for each panel. Ensuring that every figure caption is self-contained and explicitly describes the data in each sub-panel will significantly enhance the paper's accessibility. Addressing these structural and stylistic inconsistencies is essential before the paper can be considered for acceptance.

Action items.

- [**writing**] The manuscript presents a comprehensive technical contribution, but the writing quality is currently compromised by significant structural redundancy and inconsistent formatting that impedes readability. The most critical issue is the duplication of major sections. Specifically, the "Conclusion and Future Works" and "Theoretical Analysis" sections appear in multiple chunks (e.g., e002 and e003) with slight variations in phrasing. This suggests a compilation error or a failure to merge draft vers